

(iii) Use your answer in (b)(ii) to determine the amount  $n$  of gas in the sample.

$$n = \dots\dots\dots \text{mol} \quad [1]$$

- (c) The root-mean-square (r.m.s.) speed of the molecules of the gas is  $1900 \text{ ms}^{-1}$  when  $pV$  is equal to  $270 \text{ J}$ .

Determine the mass, in  $u$ , of one molecule of the gas, where  $u$  is the unified atomic mass unit.

$$\text{mass} = \dots\dots\dots u \quad [4]$$

[Total: 10]

